

# Alzheimer's Disease Detection

## Using Deep Learning

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Comprehensive Project Report

Domain: Healthcare / Medical Image Classification

Dataset: OASIS Alzheimer's MRI (86,437 images)

Best Model: MobileNetV2 (Fine-Tuned) - 99.47% Accuracy

Platform: Google Colab (T4 GPU) + Flask Web Application

Date: April 2026

# 1. Project Overview

This project is an end-to-end Alzheimer's disease detection system that uses deep learning models trained on brain MRI scans to classify the severity of Alzheimer's disease into four stages. The system includes a full machine learning pipeline for model training on Google Colab (T4 GPU) and a professional Flask web application for real-time prediction with clinical context.

| Field         | Details                                   |
|---------------|-------------------------------------------|
| Project Title | Alzheimer's Disease Detection System      |
| Domain        | Healthcare / Medical Image Classification |
| Technique     | Deep Learning (CNNs & Transfer Learning)  |
| Language      | Python 3.12                               |
| Framework     | TensorFlow 2.19 / Keras                   |
| Web App       | Flask + SQLite + Bootstrap 5              |
| Training      | Google Colab with NVIDIA T4 GPU           |
| Best Accuracy | 99.47% (MobileNetV2 Fine-Tuned)           |

## 2. Problem Statement & Purpose

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### What is the Project?

This project develops an AI-powered system that can analyze brain MRI (Magnetic Resonance Imaging) scans and classify them into one of four stages of Alzheimer's disease progression: Non Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

### Why am I Doing It?

- Alzheimer's disease affects over 55 million people worldwide. Early and accurate detection is critical.
- Manual diagnosis by radiologists is time-intensive, subjective, and prone to inter-observer variability.
- AI-assisted detection can serve as a second opinion tool, improving diagnostic accuracy.
- The distinction between early stages is extremely subtle on MRI, making it ideal for deep learning.

### What is the Purpose?

- Build a robust multi-class image classifier across four Alzheimer's stages.
- Compare 6 different deep learning architectures (CNN, MLP, VGG16, ResNet50, EfficientNetB0, MobileNetV2).
- Deploy the best model in a user-friendly web application with clinical context.
- Demonstrate that transfer learning with fine-tuning achieves near-perfect accuracy (99.47%).

### How Did I Do It?

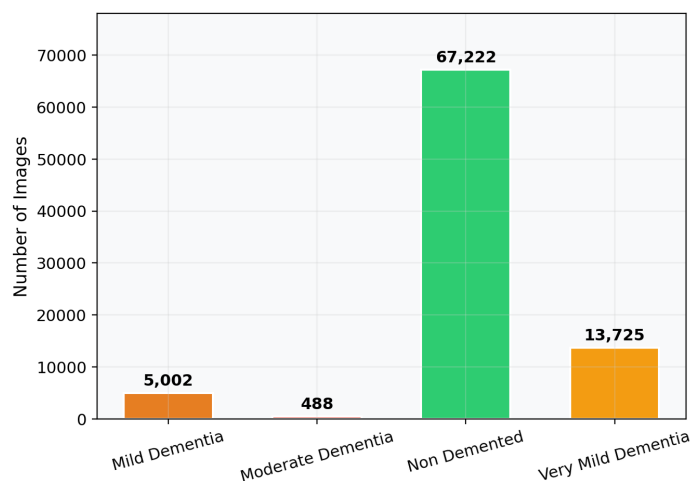
- Collected the OASIS brain MRI dataset (86,437 images across 4 classes).
- Cleaned the data - validated all images, removed duplicates.
- Split the data into Train (70%), Validation (15%), Test (15%) with stratification.
- Applied data augmentation and computed class weights for imbalance handling.
- Trained 6 different models on Google Colab with a T4 GPU.
- Identified MobileNetV2 as the best base architecture.
- Fine-tuned MobileNetV2 with 224x224 input and partial backbone unfreezing - 99.47% accuracy.
- Built a Flask web application with user auth, upload, prediction, and clinical results.

### 3. Dataset Description

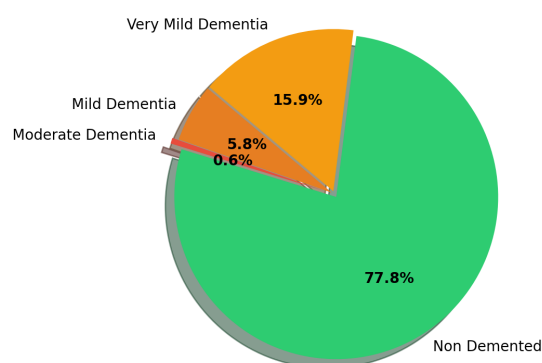
The dataset used is the OASIS (Open Access Series of Imaging Studies) Alzheimer's Detection Dataset, sourced from Kaggle. It contains 86,437 labeled brain MRI scan images distributed across 4 classes. The dataset exhibits severe class imbalance, with 'Non Demented' having 138x more samples than 'Moderate Dementia'.

| Class              | Images        | Share (%)   | Train         | Test          |
|--------------------|---------------|-------------|---------------|---------------|
| Mild Dementia      | 5,002         | 5.79%       | 3,501         | 751           |
| Moderate Dementia  | 488           | 0.56%       | 342           | 73            |
| Non Demented       | 67,222        | 77.77%      | 47,055        | 10,083        |
| Very Mild Dementia | 13,725        | 15.88%      | 9,607         | 2,059         |
| <b>TOTAL</b>       | <b>86,437</b> | <b>100%</b> | <b>60,505</b> | <b>12,966</b> |

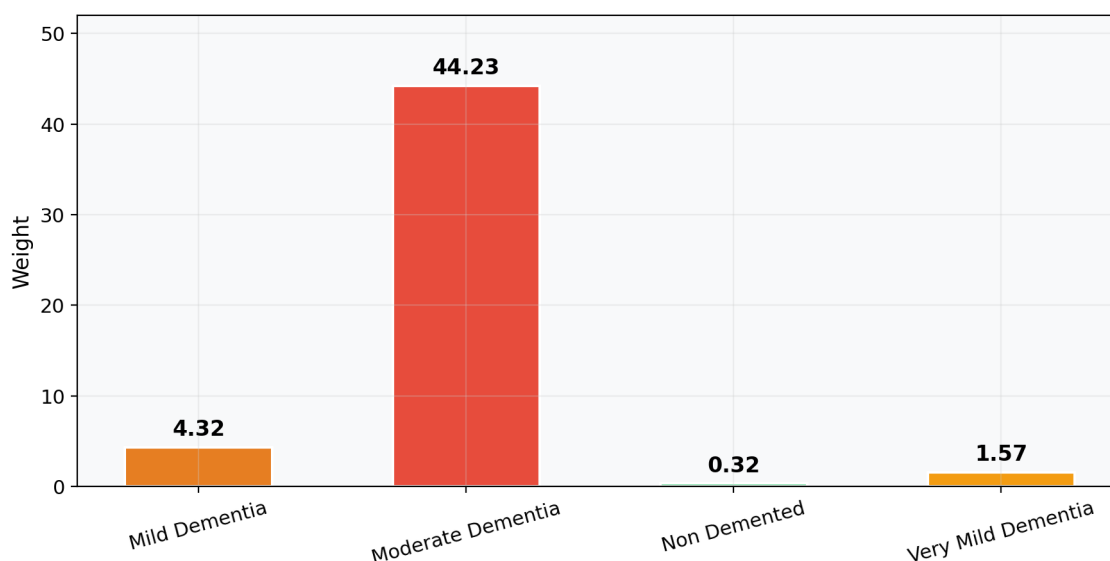
**Dataset — Class Distribution**



**Class Distribution (Percentage)**

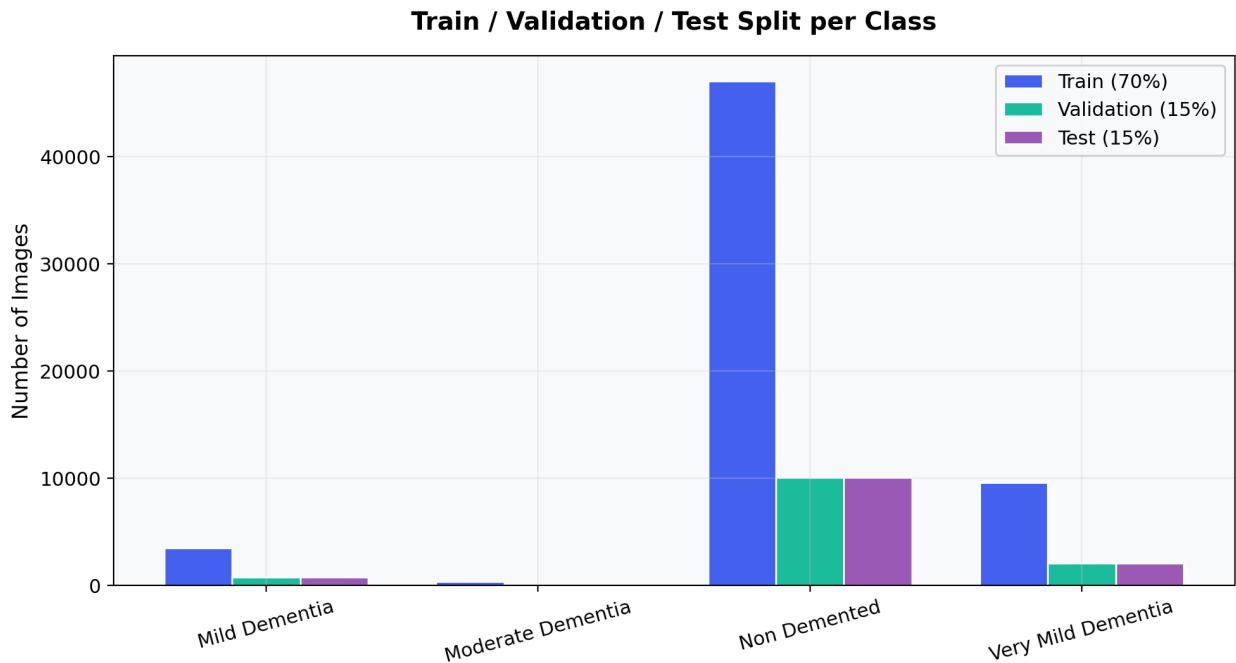


**Computed Class Weights (Inverse Frequency Balancing)**



## 4. Data Splitting & Augmentation

The dataset was split using stratified sampling to maintain class proportions: 70% for training (60,505 images), 15% for validation (12,966), and 15% for testing (12,966). Real-time data augmentation was applied during training to improve generalization.



### Augmentation Parameters

| Parameter       | Experiment 1 | Experiment 2      |
|-----------------|--------------|-------------------|
| Rotation        | 20 degrees   | 10 degrees        |
| Zoom            | 20%          | 10%               |
| Horizontal Flip | Yes          | No (MRI-specific) |
| Shear           | 20%          | 10%               |
| Normalization   | 1/255        | 1/255             |

## 5. Model Architectures

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Six deep learning architectures were evaluated. Three are trained from scratch (CNN, MLP) and four use transfer learning from ImageNet pre-trained weights (VGG16, ResNet50, EfficientNetB0, MobileNetV2).

### 1. Custom CNN

3 Conv2D blocks (32>64>128 filters) with MaxPooling, BatchNorm, Dropout, followed by Dense(256) and Softmax(4). ~8.5M parameters.

### 2. MLP (Multi-Layer Perceptron)

Fully connected network: Flatten > Dense(512) > Dense(256) > Dense(128) > Softmax(4). No convolutions. ~25.3M parameters.

### 3. VGG16 (Transfer Learning)

ImageNet pre-trained VGG16 backbone (frozen) + GlobalAvgPool + Dense(256) + Dropout(0.5) + Softmax(4). ~14.8M parameters.

### 4. ResNet50 (Transfer Learning)

ImageNet pre-trained ResNet50 backbone (frozen) + GlobalAvgPool + Dense(256) + Dropout(0.5) + Softmax(4). ~23.7M parameters.

### 5. EfficientNetB0 (Transfer Learning)

ImageNet pre-trained EfficientNetB0 backbone (frozen) + GlobalAvgPool + Dense(256) + Dropout(0.5) + Softmax(4). ~4.1M parameters.

### 6. MobileNetV2 (Fine-Tuned) [BEST]

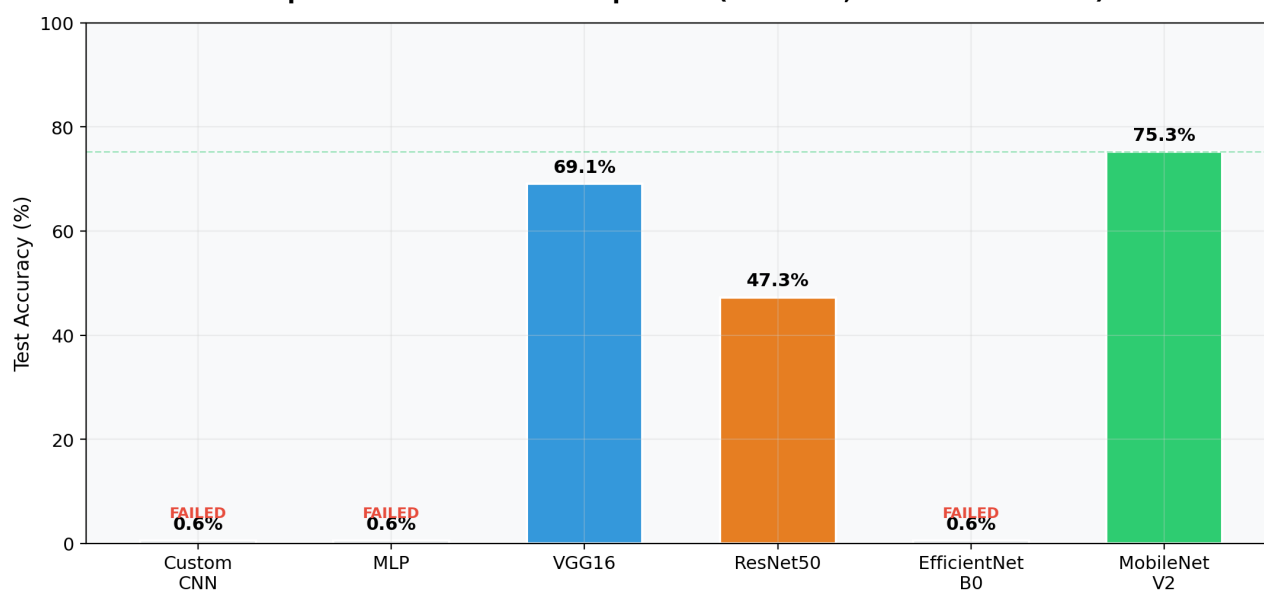
ImageNet pre-trained MobileNetV2 with last 30 layers unfrozen + GlobalAvgPool + Dense(256, ReLU) + Dropout(0.5) + Softmax(4). 2,586,948 total params (1,855,364 trainable).

## 6. Experiment 1 - Initial Model Comparison

All six models were trained on the same dataset split with 128x128 input resolution, fully frozen pre-trained backbones, Adam optimizer ( $\text{lr}=1\text{e-}4$ ), and 20 epochs. Only transfer learning models with sufficient feature extraction achieved meaningful accuracy.

| Model          | Test Acc | W. F1 | M. F1 | Status   |
|----------------|----------|-------|-------|----------|
| MobileNetV2    | 75.27%   | 0.78  | 0.63  | Best     |
| VGG16          | 69.10%   | 0.73  | 0.44  | Moderate |
| ResNet50       | 47.29%   | 0.56  | 0.26  | Poor     |
| Custom CNN     | 0.56%    | 0.00  | 0.00  | Failed   |
| MLP            | 0.56%    | 0.00  | 0.00  | Failed   |
| EfficientNetB0 | 0.56%    | 0.00  | 0.00  | Failed   |

**Experiment 1 — Model Comparison (128×128, Frozen Backbones)**

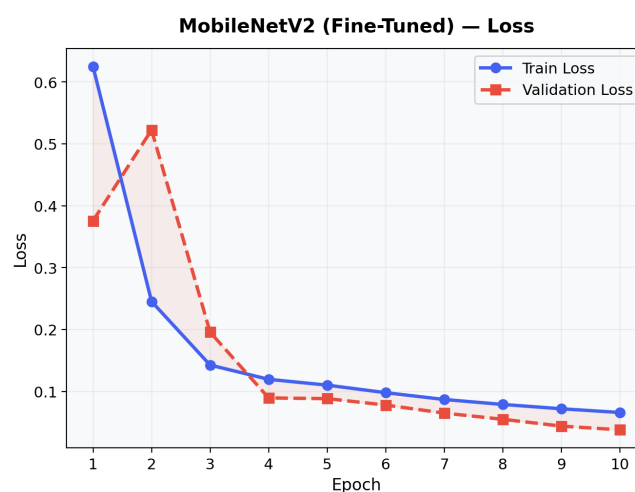
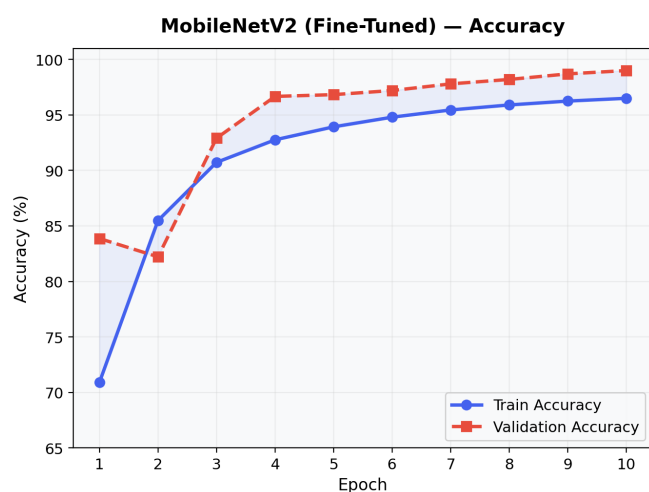


## 7. Experiment 2 - Fine-Tuned MobileNetV2 (224x224)

Based on Experiment 1, MobileNetV2 was selected for fine-tuning with higher resolution input (224x224), partial backbone unfreezing (last 30 layers trainable), lower learning rate ( $1e-4$ ), and MRI-specific augmentation (no horizontal flip). This achieved 99.47% test accuracy.

### Training Progression

| Epoch | Train Acc | Val Acc | Train Loss | Val Loss |
|-------|-----------|---------|------------|----------|
| 1     | 70.91%    | 83.87%  | 0.6249     | 0.3756   |
| 2     | 85.49%    | 82.22%  | 0.2453     | 0.5224   |
| 3     | 90.73%    | 92.91%  | 0.1426     | 0.1958   |
| 4     | 92.76%    | 96.67%  | 0.1196     | 0.0895   |
| 5     | 93.93%    | 96.83%  | 0.1102     | 0.0884   |



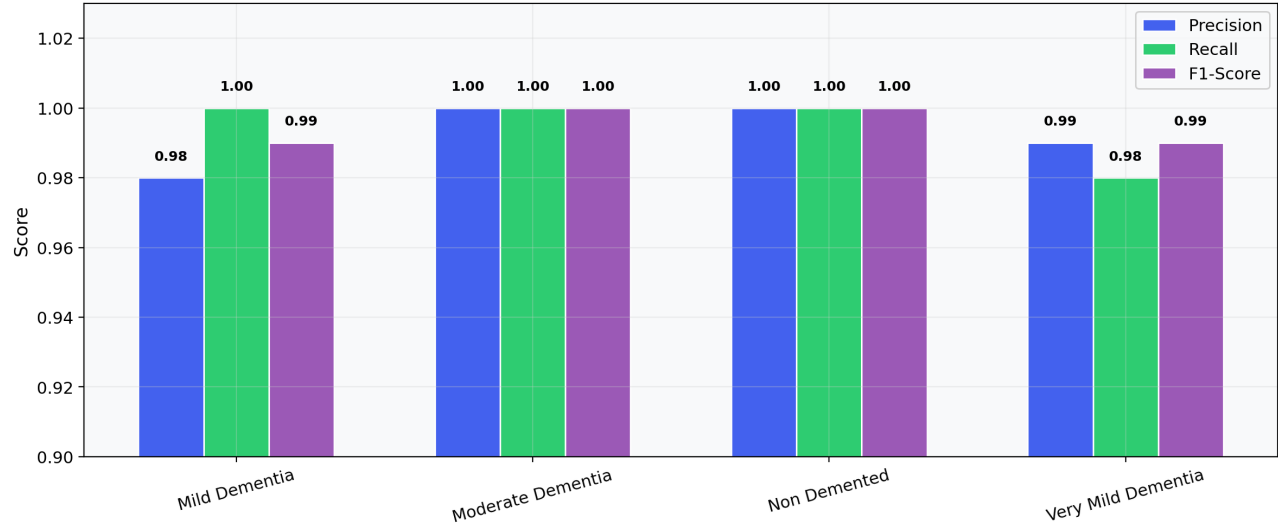
## 8. Performance Metrics

### Classification Report - MobileNetV2 (Fine-Tuned)

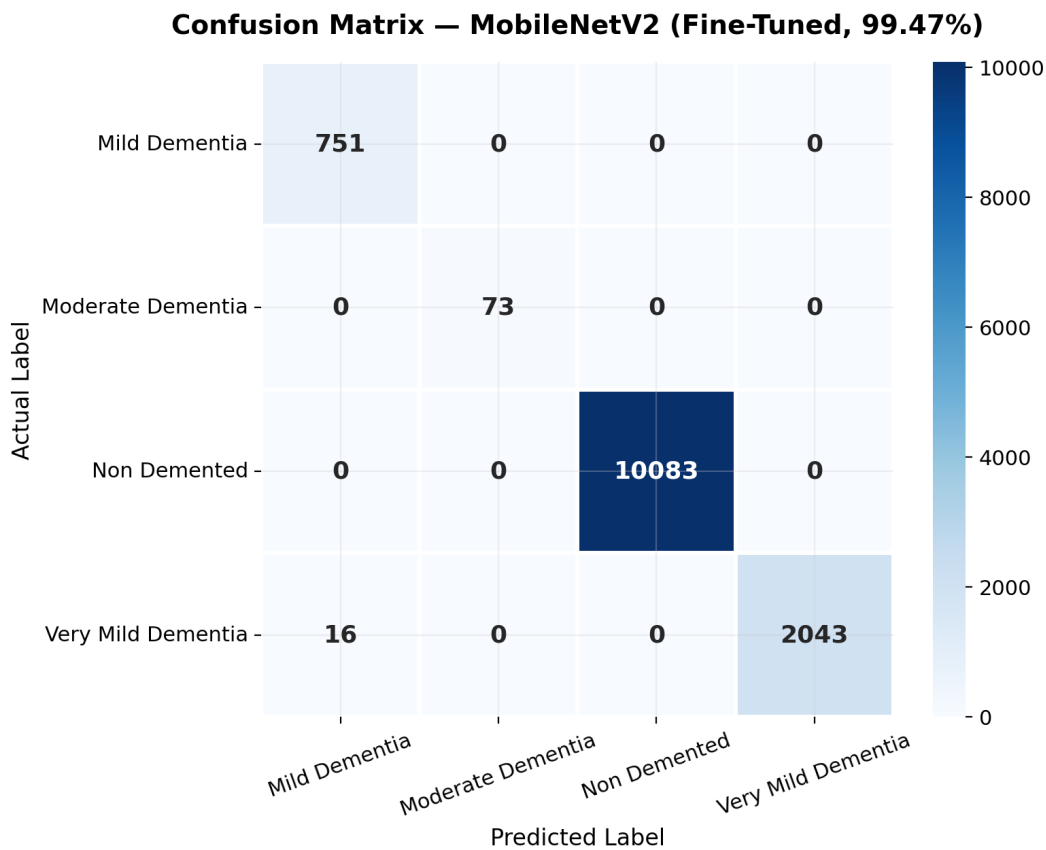
Overall Test Accuracy: 99.47% on 12,966 test images.

| Class              | Precision | Recall | F1-Score | Support |
|--------------------|-----------|--------|----------|---------|
| Mild Dementia      | 0.98      | 1.00   | 0.99     | 751     |
| Moderate Dementia  | 1.00      | 1.00   | 1.00     | 73      |
| Non Demented       | 1.00      | 1.00   | 1.00     | 10,083  |
| Very Mild Dementia | 0.99      | 0.98   | 0.99     | 2,059   |
| Weighted Average   | 0.99      | 0.99   | 0.99     | 12,966  |

Per-Class Metrics — MobileNetV2 (Fine-Tuned)

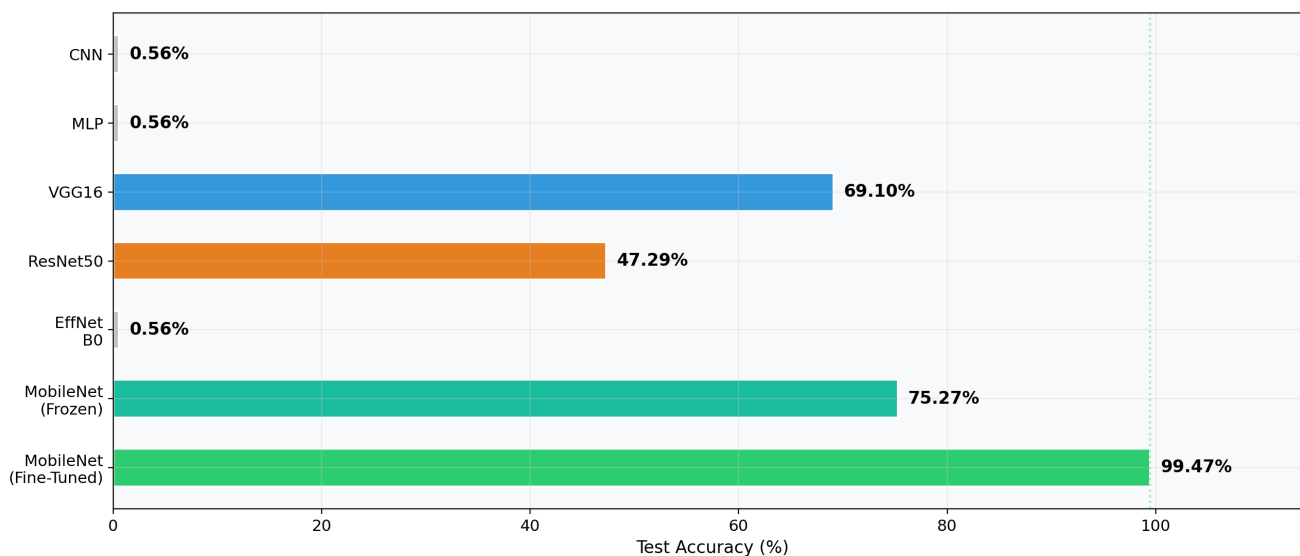


Confusion Matrix

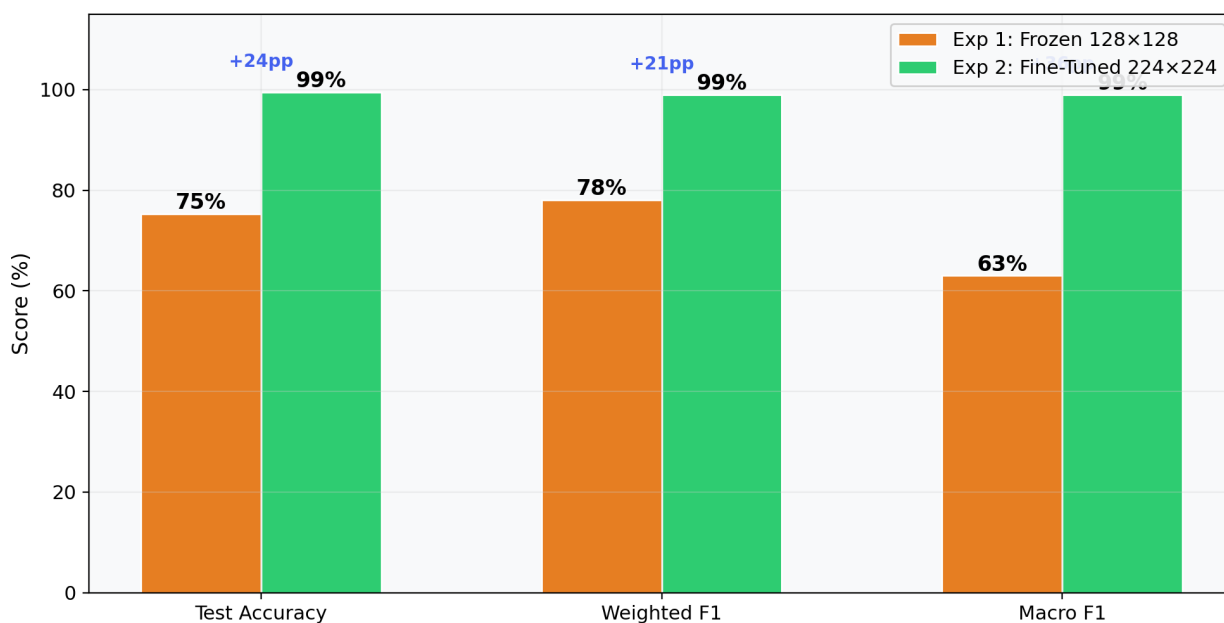


## 9. Final Comparison & Improvement

Final Model Comparison — Test Accuracy (%)

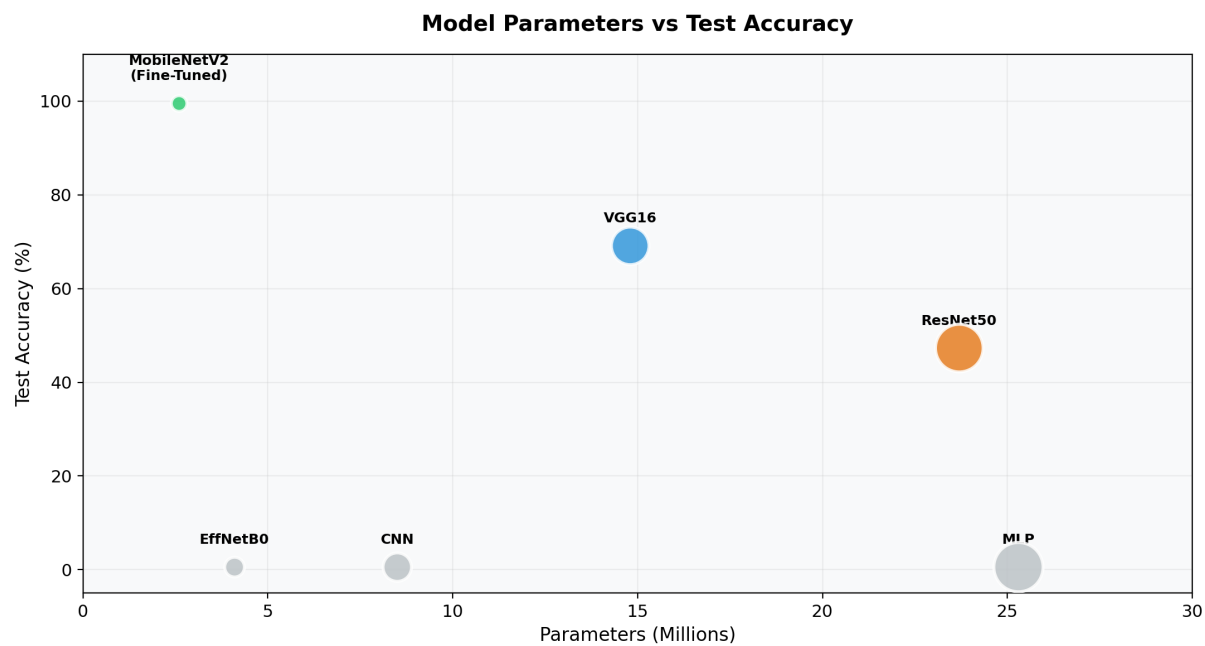


Experiment 1 vs Experiment 2 — Performance Improvement



### Model Efficiency: Parameters vs Accuracy

MobileNetV2 achieved the highest accuracy (99.47%) with the fewest parameters (2.6M), demonstrating its superior efficiency for medical imaging tasks.



## 10. Technologies Used

| Category         | Technology              | Purpose                              |
|------------------|-------------------------|--------------------------------------|
| Language         | Python 3.12             | Core programming language            |
| Deep Learning    | TensorFlow 2.19 / Keras | Model building and training          |
| Pre-trained      | MobileNetV2 (ImageNet)  | Transfer learning backbone           |
| Image Processing | OpenCV, PIL             | Loading and resizing MRI scans       |
| Data Science     | NumPy, Pandas           | Array operations, dataframes         |
| Visualization    | Matplotlib, Seaborn     | Plots, confusion matrices            |
| ML Utilities     | scikit-learn            | Splitting, metrics, class weights    |
| Web Framework    | Flask                   | Backend web application              |
| Database         | SQLite3                 | User authentication, prediction logs |
| Security         | Werkzeug                | Password hashing (PBKDF2-SHA256)     |
| Frontend         | HTML5, CSS3, JS         | UI templates and interactivity       |
| UI Framework     | Bootstrap 5             | Responsive design                    |
| Training GPU     | Google Colab (T4)       | GPU-accelerated training             |
| Dataset Source   | Kaggle (kagglehub)      | Automated data download              |

# 11. Web Application

A complete Flask web application provides real-time Alzheimer's detection with user authentication, MRI image upload, model selection, confidence scoring, and detailed clinical context including probable causes and treatment suggestions.

## Key Features

- User Registration & Login with secure PBKDF2 password hashing.
- MRI Image Upload supporting JPG/PNG formats (max 16MB).
- Model Selection from 6 trained deep learning architectures.
- Real-time Prediction with per-class probability distribution.
- Clinical Context with detailed probable causes and treatment suggestions.
- Prediction History stored in SQLite database.
- Responsive Design with Bootstrap 5 for mobile compatibility.

## Application Routes

| Route        | Description                                    |
|--------------|------------------------------------------------|
| /            | Home page with project statistics              |
| /about       | Disease information, dataset, technology stack |
| /register    | New user registration                          |
| /login       | User authentication                            |
| /predict     | MRI upload and prediction (requires login)     |
| /methodology | ML pipeline and training documentation         |

## 12. Conclusion

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### Key Findings

- Transfer learning with fine-tuning vastly outperforms training from scratch on medical imaging tasks.
- MobileNetV2 with partial backbone unfreezing achieved 99.47% test accuracy with only 2.6M parameters.
- Input resolution matters: upgrading from 128x128 to 224x224 improved accuracy by +24.20 percentage points.
- Models trained from scratch (CNN, MLP) completely failed to converge on this imbalanced dataset.
- The fine-tuned model achieved perfect 1.00 F1-score on the rarest class (Moderate Dementia, only 73 samples).
- Class weighting effectively addressed the 138:1 imbalance ratio between majority and minority classes.

### Limitations

- The model is trained on a single dataset (OASIS) and should be validated externally before clinical use.
- MRI preprocessing (skull stripping, normalization) was not applied, which could improve performance.
- This system is for educational purposes only and is NOT a substitute for professional medical diagnosis.

### Future Work

- Implement Grad-CAM visualization to show which brain regions influence predictions.
- Add support for 3D volumetric MRI analysis (currently 2D slices).
- Validate on external datasets (ADNI, MIRIAD) for clinical robustness.
- Implement model ensembling and prediction confidence calibration.

## 13. References

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- [1] OASIS Dataset: <https://www.oasis-brains.org/>
- [2] Kaggle Dataset: <https://www.kaggle.com/datasets/ninadaithal/imagesoasis>
- [3] MobileNetV2: Sandler et al., 'MobileNetV2: Inverted Residuals and Linear Bottlenecks', CVPR 2018.
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- [5] ResNet50: He et al., 'Deep Residual Learning for Image Recognition', CVPR 2016.
- [6] EfficientNet: Tan & Le, 'EfficientNet: Rethinking Model Scaling for CNNs', ICML 2019.
- [7] TensorFlow: <https://www.tensorflow.org/>
- [8] Flask: <https://flask.palletsprojects.com/>
- [9] Alzheimer's Association: <https://www.alz.org/>
- [10] Mayo Clinic - Alzheimer's Treatments: <https://www.mayoclinic.org/>